



Customer Segmentation Analysis for Strategic Growth

Identifying and Retaining High-Value Retail Customers

Project Overview & Methodology



Transitioning Data to Insights

The strategic imperative is to leverage clean behavioral insights to convert standard shoppers into loyal advocates. We refined 541,909 raw records down to 391,297 valid product transactions by removing missing IDs and non-product codes.

Using RFM analysis (Recency, Frequency, Monetary), we established baseline behavioral profiles. To address severe skewness in retail distributions, we applied log transformation and K-Means clustering, enabling highly targeted, segment-specific retention models.

Data Quality Filtering

391,297

VALID TRANSACTIONS

Data Cleanliness Rate

72.2%

541,909 Raw Records

Filtered Dataset

Behavioral Customer Segments



K-Means clustering reveals three distinct customer segments with varying engagement levels.



Champions

1,014 Customers

High-value shoppers with high frequency (256.04 average) and massive monetary value (£6,254.51 average) who engaged recently (16.10 days).



Potential Loyalists

1,842 Customers

The reliable middle tier demonstrating consistent purchasing behavior (59.44 average frequency, £1,077.76 average spend).



At-Risk / Lapsed

1,478 Customers

Re-engagement targets characterized by long inactivity (178.64 days average recency) and lower overall spend (£289.55 average).



Strategic Recommendations

Targeted marketing strategies will drive VIP retention, cross-selling growth, and re-engagement.

- **Retention (Champions):** Implement an exclusive VIP loyalty program offering early access to new product launches to sustain high transaction frequency.
- **Growth (Potential Loyalists):** Deploy cross-selling tactics and starter bundles combining utility items with top products like Ceramic Jars to increase lifetime value.
- **Re-engagement (At-Risk):** Launch targeted win-back email campaigns featuring deep discounts on low-friction items like balms and purses to incentivize a return purchase.

Align marketing campaigns with behavior to maximize lifetime value and ROI.

Model Limitations & Deployment Risks



Real-world constraints require careful contextualization of the segmentation model before deployment.

Class Imbalance: The massive At-Risk majority versus the dense Champions minority reflects real-world Pareto distributions, which can bias marketing resource allocation if unmanaged.

Temporal Overfitting: Anchoring the snapshot date to the maximum invoice date risks look-ahead bias if deployed directly on live streaming data.

Geographic Skew: The dataset is predominantly driven by UK wholesale and retail transactions. Direct application to international markets requires country-wise normalization to avoid misinterpreting local shopping habits.